

Full Length Article

Ionic liquid incorporation in zeolitic imidazolate framework-3 for improved CO₂ separation: A computational approachAnoopa Thomas, Muthuramalingam Prakash^{*}

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ARTICLE INFO

Keywords:

CO₂ capture

ZIF-3

Ionic liquids

Composites

GCMC simulations

ABSTRACT

Selective adsorption of gases from various mixtures of industrial importance has become a major concern. Improving the selectivity of the existing material by chemical/physical modification can solve this problem to an extent. In this regard, we incorporated ILs into the pores of well-known porous material, zeolitic imidazolate framework (ZIF). ZIF-3 has shown good selectivity for CO₂ but their modification with ionic liquid (IL) has not yet explored. We studied 1-butyl-3-methylimidazolium ([BMIM]⁺) cation with seven different anions inside the confinement of ZIF-3 and their effect on the separation of CO₂. Different orientations are checked and the stable one is selected for advanced computational studies. Pristine ZIF-3 and IL impregnated ZIF-3 shows good adsorption potential for CO₂. The incorporation of ILs in ZIF-3 significantly enhances the CO₂ capture from CO₂/N₂ and CO₂/CH₄ gaseous mixture. Whereas it is interesting that the adsorption of methane from CH₄/N₂ highly depends on the nature of IL (i.e. anion) inside the ZIF-3. Adsorption energies and heat of adsorption gives the strength of interaction between ILs, ZIF-3 and guest gaseous molecules. The ILs@ZIF-3 composites can perform better than well-known IL@ZIF-8 composite at higher pressure. The investigation of IL incorporation to various ZIFs can offer better CO₂ selective material than the existing ones in the near future. Our combined density functional theory (DFT) and grand canonical Monte Carlo (GCMC) simulations are used to get more insights on the interfacial interactions between the ILs-ZIF structures and how these molecular level interactions can be helpful in the gas sensor applications and industrial flue gas separations.

1. Introduction

The emission of CO₂ from power plants and fossil fuels is of major concern, as it enhances the global warming and resulting climate changes [1]. The development of new technologies for the capture of the toxic gases from the atmosphere is gaining the attention of the scientific world. The selective separation and long term storage of gases have many barriers to overcome. Feasible carbon capture and sequestration (CCS) techniques should be commercially implemented to reduce the environmental impact [2]. The current carbon capture techniques have many disadvantages such as high energy consumption and less efficiency. Many materials which showed good competence are in developmental stage and need to overcome several barriers before commercialization [3]. Adsorption and absorption are the two main techniques to effectively capture CO₂ from various sources [4]. Among the adsorption techniques, membrane based separation techniques are emerging as potential tools for gas separation [5,6]. The design of highly selective membranes and their implementation in real-time CO₂ capture

is in the preliminary stage.

Porous materials are one of the main focuses of researchers in the field of gas handling. They can be combined with many superior materials to form membranes [7–9]. Metal-organic frameworks (MOFs) and their sub-class zeolitic imidazolate frameworks (ZIFs) have gained tremendous attention in the field of gas capture [10,11]. Especially, they are prominent in CO₂ capture from flue gas mixtures. Different ZIFs such as ZIF-7, -8, -10, -68, -69, -70, -95, -100 etc. are used to study the adsorption and separation of different gases from industrial flue gas [12–16]. Seda Keskin and co-workers studied different ZIFs and their gas mixture selectivity. They found that ZIFs can perform better than many of the zeolites and MOFs in CO₂ capture [17]. But many studies have focused on specific ZIFs for gas separation, especially ZIF-8 [18–20].

Ionic liquid (IL) is a green solvent and is a leading material in gas handling from the last decade [21,22]. By itself, ILs have many disadvantages such as high viscosity and toxicity. Supporting them over different membranes can give a solution to such issues [23]. Supported IL membranes (SILMs) are a novel class which combine the advantages

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<https://doi.org/10.1016/j.apsusc.2021.150173>

Received 8 April 2021; Received in revised form 27 April 2021; Accepted 19 May 2021

Available online 24 May 2021

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of chemical and physical adsorption with the ease of separation [23–25]. ILs can be confined into various materials such as carbon nanotubes (CNT), porous silica, polymers, MOFs etc. [26,27]. ILs can be confined into the pores of ZIFs and these composite materials can act as good selective membrane for CO₂. Confined ILs can selectively separate CO₂ from other mixtures [28,29]. Many groups have investigated the IL/ZIF-8 composites [30–32]. ZIF-8 is very compatible with ILs and this can enhance the CO₂ selectivity [32,33].

In this study, we have selected ZIF-3 as the supporting material for ILs [10]. ZIF-3 has DFT topology with imidazolium linker whereas ZIF-8 has SOD topology with methylimidazolium linker. It has a channel-type pore unlike ZIF-8 which has cage like pore. ZIF-3 has higher surface area and pore volume than ZIF-8 [34]. But there are only a few studies on ZIF-3 in context of gas capture. Zheng et al studied the effect of framework charges on the CO₂ uptake of ZIF-3 and various other MOFs using force field based GCMC simulations. They found that the CO₂ uptake in MOFs can be pore size dependent and is adsorbed in small pores of ZIF-69 [35]. Han et. al identified ZIF-3 as potential material for H₂ adsorption. They also found that H₂ adsorption depends on surface area, not on the topology of ZIFs [34]. Guo et. al investigated ZIF-3 along with other ZIFs and identified it as an efficient material for CH₄/H₂ separations [36]. Seda Keskin investigated ZIF-3 and ZIF-10 for their selective gas separation (CO₂/H₂, CO₂/CH₄ and CH₄/H₂) using GCMC and molecular dynamics (MD) simulations [37]. Pantatosaki et. al compared the behavior of ZIF-3 and ZIF-8 structures with sorbent molecules using experimental and MD techniques [38]. Yilmaz et. al screened different ZIFs using computational techniques for CO₂ separation and among them ZIF-3 also shows promising results [17]. Komiha and co-workers used DFT techniques to investigate the CO₂ adsorption in ZIFs. They found that CO₂ is physisorbed on these ZIFs and are governed by hydrogen bonding (H-bonding) and π -stacking, which can induce structural changes in ZIFs [39].

Even though there are few studies about gas capture and separation in ZIF-3, there is no investigation on IL impregnated ZIF-3 (i.e. ILs@ZIF-3) based composite materials. We have previously studied different ILs inside the confinement of ZIF-8 [32,33,40,41]. Here, we took imidazolium based [BMIM] cation with various anions and they were impregnated into the pores of ZIF-3. These composites are investigated using periodic DFT calculations for finding the stability of ILs in the confinement. Further to get more understanding on the adsorption capacity and gas selectivity, we used force field based GCMC simulations. Computational approach can be useful in understanding the interactions of small molecules in MOFs [42]. The computational approach can give detailed insights about how these ILs behave at the ZIF-3 surface and the behavior of incoming gas molecules at the solid-liquid interface. In addition, the effect of anion in the ion-pair and the orientation of different ILs at the ZIF-3 is studied. This is the first report on investigation of various ILs impregnated ZIF-3 material for the selective CO₂ capture application. The molecular level understanding of such systems can be helpful in studying different ILs@ZIFs composites in the field of gas capture and separation.

2. Computational methodology

The optimized ILs [32] are inserted into the pores of ZIF-3 crystal structure. For each IL, three different orientations are checked. All the DFT based calculations are done in CP2K 5.1 software using quickstep package [43]. CELLOPT method was used to relax the both pristine ZIF-3 and ILs@ZIF-3 composite structures. This helps to understand the structural evaluation (i.e. volume changes) of ZIF-3 while incorporating ILs [41]. Periodic boundary condition (PBC) was used for all calculations. PBE-D2 (Perdew-Burke-Ernzerhof[44] method with Grimme's D2 [45] semi-empirical dispersion correction) was used for the calculations. From our previous study, we found that DFT-D2 method can predict ILs@ZIFs composites accurately [41]. Hence, we employed the PBE-D2 method for all composites. Gaussian plane wave (GPW) [46] basis set

scheme was used with triple-zeta valence polarized (MOLOPT-TZVP) [47] basis set for atoms except Zn and B. For these two atoms, double-zeta valence polarized (MOLOPT-DZVP) basis set was used [47]. 450 Ry plane wave cutoff was used and Goedecker-Teter-Hutter [48] (GTH) pseudo potentials were used for all atoms. The strength of interaction between ILs and ZIF-3 is calculated from the energy of adsorption (E_{ads}) as given in Eq. (1),

$$E_{ads} = E_{Total} - (E_{IL} + E_{ZIF-3}) \quad (1)$$

E_{Total} is the total energy of the composite, E_{IL} and E_{ZIF-3} are the energies of IL and ZIF-3 respectively. The E_{ads} is calculated for all the three different orientations and the one with maximum E_{ads} is selected for the GCMC simulations.

GCMC simulations are used to study the gas adsorption and selectivity at the designed composites. All GCMC calculations are done using RASPA software [49]. In our previous work, we inferred that Dreiding force field [50] is better in predicting gas adsorption at IL@ZIF-8 composites compared to other force fields [32]. Hence we selected the same force field for all the framework atoms. The parameters for the gas molecules (CO₂, N₂ and CH₄) are taken from TraPPE force field [51]. CO₂ and N₂ are considered as three site model whereas CH₄ is considered as single sphere. The host-guest interactions are defined using 12–6 Lennard-Jones (LJ) parameters. Cross-atomic parameters for the force field are defined using Lorentz-Berthelot mixing rules [52]. Charges for the framework atoms are calculated using charge equilibration method. Previous report suggests that the charge equilibration method can predict comparable results with that of predicted by Mulliken population analysis [33]. 1×10^5 steps were used for initialization and 1×10^6 steps for equilibration. Rotation, translational, reinsertion and swap moves were used for single component calculations and identity exchange was added for binary gas mixture calculations. All calculations are carried out at 0.1 to 10 bar pressure and 298 K. The selectivity from equimolar binary gas mixture is calculated using the following Eq. (2),

$$S_{a/b} = \frac{(N_a/N_b)}{(Y_a/Y_b)} \quad (2)$$

where N and Y are the amount adsorbed and mole fraction of the gases a and b, respectively. Also the heat of adsorption (ΔH_{ads}) of gas molecules at the composites at infinite dilution condition is calculated using Eq. (3) [49],

$$\Delta H_{ads} = \langle U_{hg} \rangle - \langle U_g \rangle - \langle U_h \rangle - RT \quad (3)$$

U_{hg} is the host-guest energy, U_g and U_h are the energies of guest and host respectively. R is the universal gas constant and T is the temperature (298 K).

3. Results and discussions

3.1. Geometries of ZIF-3 and ILs@ZIF-3

In order to understand the structural changes in the ZIF framework, first we optimized ZIF-3 without any ILs inside it. The structures before and after optimization are given in Fig. 1. The figure reveals that the gates of the framework are slightly narrowed after optimization (from 4.11 to 3.86 Å). The slight changes in the pore may directly affect the gas adsorption, but this can be favorable for the selective gas capture by considering their kinetic diameter. The optimized ILs are incorporated into the crystal structure of ZIF-3 and are optimized without any constraints. The composite can be represented as [BMIM]⁺[X][−]@ZIF-3. The full names and geometries of the seven anions considered in this study are given in Supporting Information Fig. S1. For each IL, three different orientations inside the confinement are studied. These different orientations are given in Supporting Information Fig. S2. The optimized geometries of all the orientations for different composites are given in

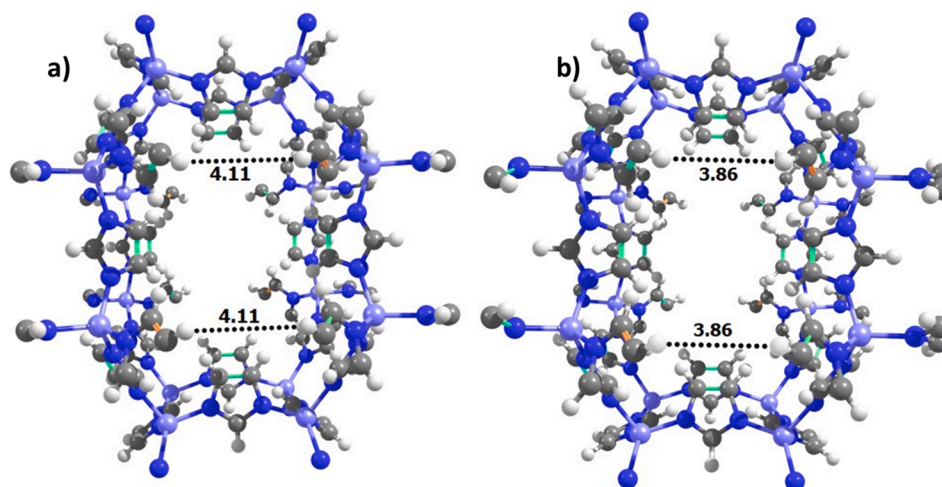


Fig. 1. The 1x1x1 unit cell of ZIF-3 a) before optimization and b) after optimization (distances are given in Å).

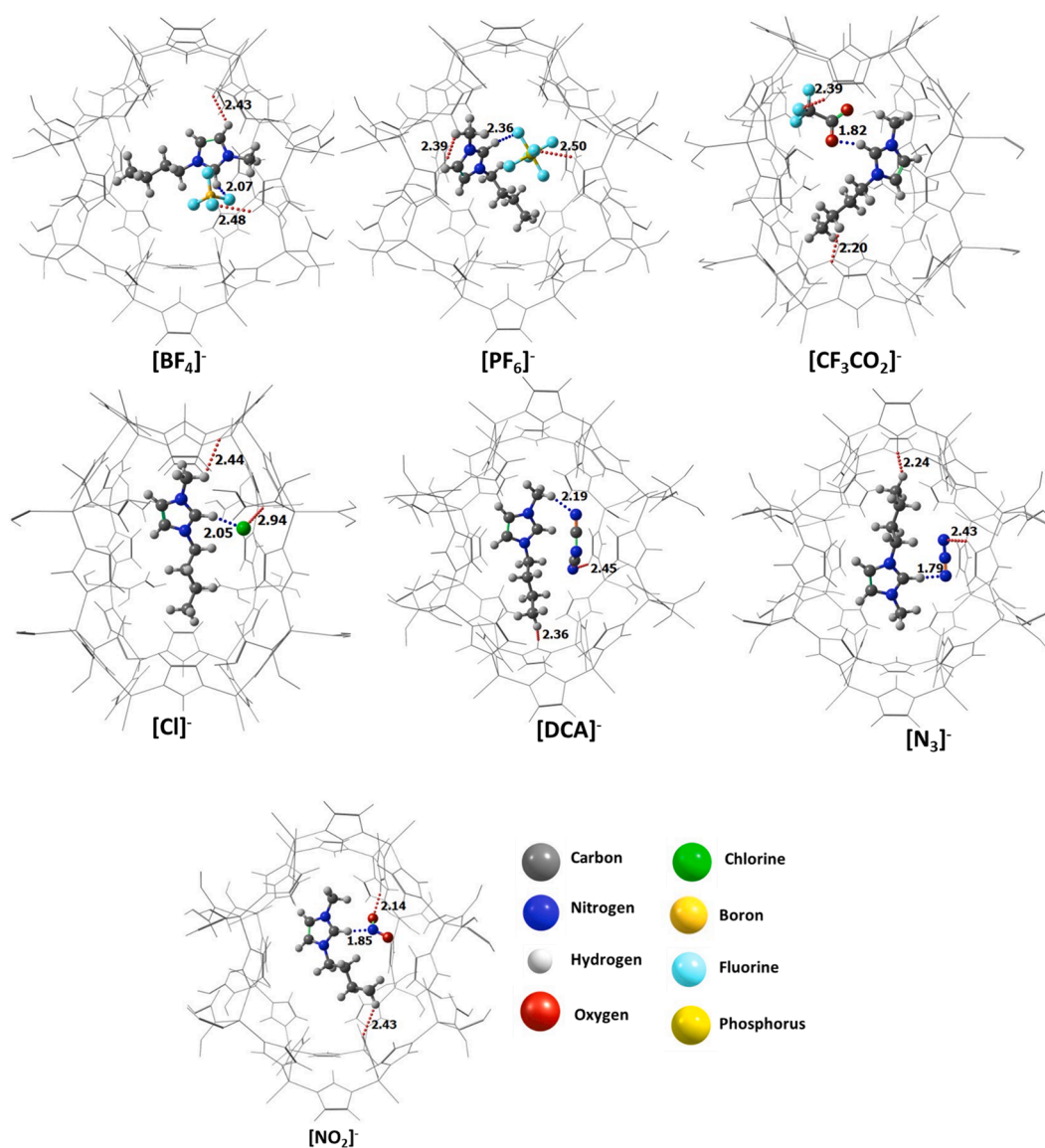


Fig. 2. Optimized geometries of different [BMIM]⁺[X]⁻@ZIF-3 (shortest interaction distances are given in Å).

Supporting Information Fig. S3. The E_{ads} of all composites in different orientations are given in Supporting Information Table S1. Also, the optimized Cartesian coordinates of all three orientations of [BMIM][BF₄]@ZIF-3 are provided in Supporting Information Table S2. This can give more clarity about the different orientations that have been checked. Adsorption energies of all the orientations are calculated and the one with highest E_{ads} values are taken for gas storage and separation purpose. In case of anions like [PF₆]⁻, [CF₃CO₂]⁻ and [NO₂]⁻, the first orientation is more stable. For [Cl]⁻, [DCA]⁻ and [N₃]⁻, the second orientation is the stable one and only for [BF₄]⁻, the third orientation is more stable. The optimized geometries of selected composites are given in Fig. 2 along with the shortest interaction distances (in Å). Some of the anions such as [NO₂]⁻, [N₃]⁻ and [CF₃CO₂]⁻ has strong H-bond with the [BMIM]⁺ cation (~1.8 Å). All the anions and their respective [BMIM]⁺ have strongly interacted with the ZIF-3 framework. So the ZIF-3 crystal structure has been expanded corresponding to the experimental crystal. The volume changes of each composite are provided in Table 1. The [BF₄]⁻ and [DCA]⁻ based composites has the highest and least volume expansion among all the composites, respectively. No composites show drastic changes in the volume after incorporation of ILs. The computed volume expansion ranges from ~ 1 to 3.5% for all the composites. This shows that all the ILs are compatible with the ZIF-3 framework and these ion-pairs at the confined environment does not affect the core structure.

The calculated E_{ads} of different ILs@ZIF-3 (the highest value in the different orientations) are provided in Table 1. [BMIM][BF₄]@ZIF-3 has the strongest interaction with the framework among all the ILs. The corresponding E_{ads} is -295.76 kcal/mol. The least E_{ads} is observed for [BMIM][N₃]@ZIF-3 which is -28.38 kcal/mol. In our previous work we found that nitrogen-based anionic ILs are not compatible with ZIF-8 framework [32]. But here at ZIF-3, all the ILs shows good compatibility and all the complexes are significantly stable. It tells that the compatibility of ILs in the framework depends on the nature of the anion and the framework. In [BMIM][BF₄]@ZIF-3, the Im⁺ (-C-H) ring of cation is interacting with the framework, which can be the main cause of the high E_{ads} value. Whereas in ZIF-8, this ring is functionalized by methyl group [41]. Thus, the E_{ads} values are significantly higher in the case of ZIF-3 structure. In ILs such as [PF₆]⁻ and [Cl]⁻, the methyl group in the cation is in direct contact with the framework. In all other four ILs ([CF₃CO₂]⁻, [DCA]⁻, [N₃]⁻, [NO₂]⁻), the alkyl chain has interacted with the framework. Meanwhile all the anions shows good contact with the framework at ~ 2.1 to 3.0 Å. All these interactions lead to good compatibility between ILs and ZIF-3. The complexes with highest E_{ads} values ([BMIM][BF₄] and [BMIM][DCA]@ZIF-3) are further analyzed with electron density analysis (EDA). The electron density maps of these two complexes are given in Supporting Information Fig. S4. Here, it is evident that there is a strong interaction of [BF₄] anion with metal center (i.e. Zn) and imidazolium linker of ZIF-3 framework. In the case of [BMIM][DCA]@ZIF-3, the [DCA] anion is directly interacting with the imidazolium linker. But there is no interaction with the metal center. So the interaction energy is reduced for [BMIM][DCA]@ZIF-3 compared

to [BMIM][BF₄]@ZIF-3. Unlike in ZIF-8, the nitrogenous anions also shows good interaction with the framework [32]. This is due to the change in organic linker and topology of ZIF-3. Organic linker in ZIF-3 is imidazolium whereas in ZIF-8 it is 2-methylimidazolium. The C₂-H part of imidazolium linker is more active in ZIF-3 and thus shows good adsorption for the both non-/nitrogenous anions. The compatibility and composite stability highly depends on the interfacial interaction between ILs and ZIF-3. Further, the impact of IL on the surface and its role in the selective gas adsorption/separation is studied.

3.2. GCMC simulations

3.2.1. Adsorption isotherms of CO₂, N₂ and CH₄@Composite

The optimized geometries of ZIF-3 and ILs@ZIF-3 are used for studying the adsorption isotherm with GCMC simulations. At first, the adsorption isotherms of gases such as CO₂, N₂ and CH₄ are calculated from single component adsorption studies. The comparison of adsorption isotherms of these gases in different composites are given in Fig. 3. The ZIF-3 as well as the composites shows similar adsorption capacity. CO₂ is adsorbed in more quantity compared to other two gases. Seda Keskin has predicted the adsorption isotherms of different gases in ZIF-3 and ZIF-10. This study reveals that CO₂ is more adsorbed than other gases (CH₄ and H₂) due to its quadruple moment [37]. The CO₂ adsorption isotherm is given in Fig. 3a. At the onset of adsorption ZIF-3 and ILs@ZIF-3 shows similar adsorption. But as the pressure increases, the isotherms of [BF₄]⁻ and [PF₆]⁻ based composites exceeds others and at very high pressure (7–10 bar), the ZIF-3 tops the isotherms of all the composites. This shortfall of the IL composites may be due to the occupation of ILs in the pore and thus incoming gas molecules are blocked from entering at high pressure. Similar observation is seen for CO₂ adsorption in ILs@ZIF-8 as well [32].

Fig. 3b and c gives the adsorption isotherms of N₂ and CH₄, respectively. They also show similar trends to that of CO₂ isotherm. At low pressures, all the complexes show similar performance but at high pressure, ZIF-3 outperforms the isotherms of the composites. Also we compared the adsorption isotherms of CO₂, N₂ and CH₄ in individual complexes. This is provided in Supporting Information Fig. S5. In all the composites, CO₂ is adsorbed in abundance compared to others. This prove that ILs@ZIF-3 can be used in CO₂ storage application. Also the CO₂ adsorption in ZIF-3 is compared with our previous results in ZIF-8 [32] and is given in Fig. S6. It was found that ZIF-3 can accommodate more CO₂ when compared to ZIF-8 framework. The CO₂ adsorption depends on the available pore, surface area and distinct functional groups. The pore volume and surface area of ZIF-3 is higher than that of ZIF-8 [34]. So the ZIF-3 framework can adsorb more CO₂ than ZIF-8. To understand the efficiency of the framework as a separation membrane, the selective adsorption of CO₂ from the binary mixture is also studied and discussed in the next section.

3.2.2. Selective adsorption from binary gas mixtures

Many earlier studies have reported the incorporation of IL into the pores of MOF/ZIF thus the enhancement in the CO₂ selectivity [53–55]. Also there has been studies on ZIF-3 for selective gas separation [35,37,39]. But there is no collective information about how IL incorporation can influence the interfacial interactions and thus affect the CO₂ selectivity at this framework. ZIF-3 and ILs@ZIF-3 act as good storage material for CO₂. But in order to understand the potential of these materials as efficient separation membranes, we studied the separation of three different gas mixtures, CO₂/N₂, CO₂/CH₄ and CH₄/N₂ at these surfaces. All the gas mixtures are taken in equimolar concentrations. Since, ZIF-3 and IL@ZIF-3 act as good storage materials for CO₂, we compared the CO₂ uptake from single component and binary mixture (CO₂/N₂) adsorption in ZIF-3 and [BMIM][BF₄]@ZIF-3. Their comparison plot is given in Supporting Information Fig. S7. It is observed that in both ZIF-3 and [BMIM][BF₄]@ZIF-3, the CO₂ is adsorbed in high quantity during single component adsorption. The pure CO₂ adsorption

Table 1

The unit cell parameters, volume change and adsorption energies of different ILs@ZIF-3 composites.

[BMIM] ⁺ [X] ⁻ @ZIF-3	a = b (Å)	c (Å)	α = β = γ (°)	Volume change (%)	E_{ads} (kcal/ mol)
ZIF-3 (crystal structure) ¹⁰	18.97	16.74	90	–	–
ZIF-3 (optimized)	19.17	16.90	90	3.11	–
[BF ₄] ⁻	19.09	17.10	90	3.44	–295.76
[PF ₆] ⁻	19.13	16.79	90	1.97	–56.16
[CF ₃ CO ₂] ⁻	19.10	16.88	90	2.75	–84.62
[Cl] ⁻	19.12	16.80	90	2.02	–49.65
[DCA] ⁻	19.10	16.70	90	1.12	–125.39
[N ₃] ⁻	19.12	16.88	90	2.43	–28.38
[NO ₂] ⁻	19.14	16.93	90	2.97	–36.41

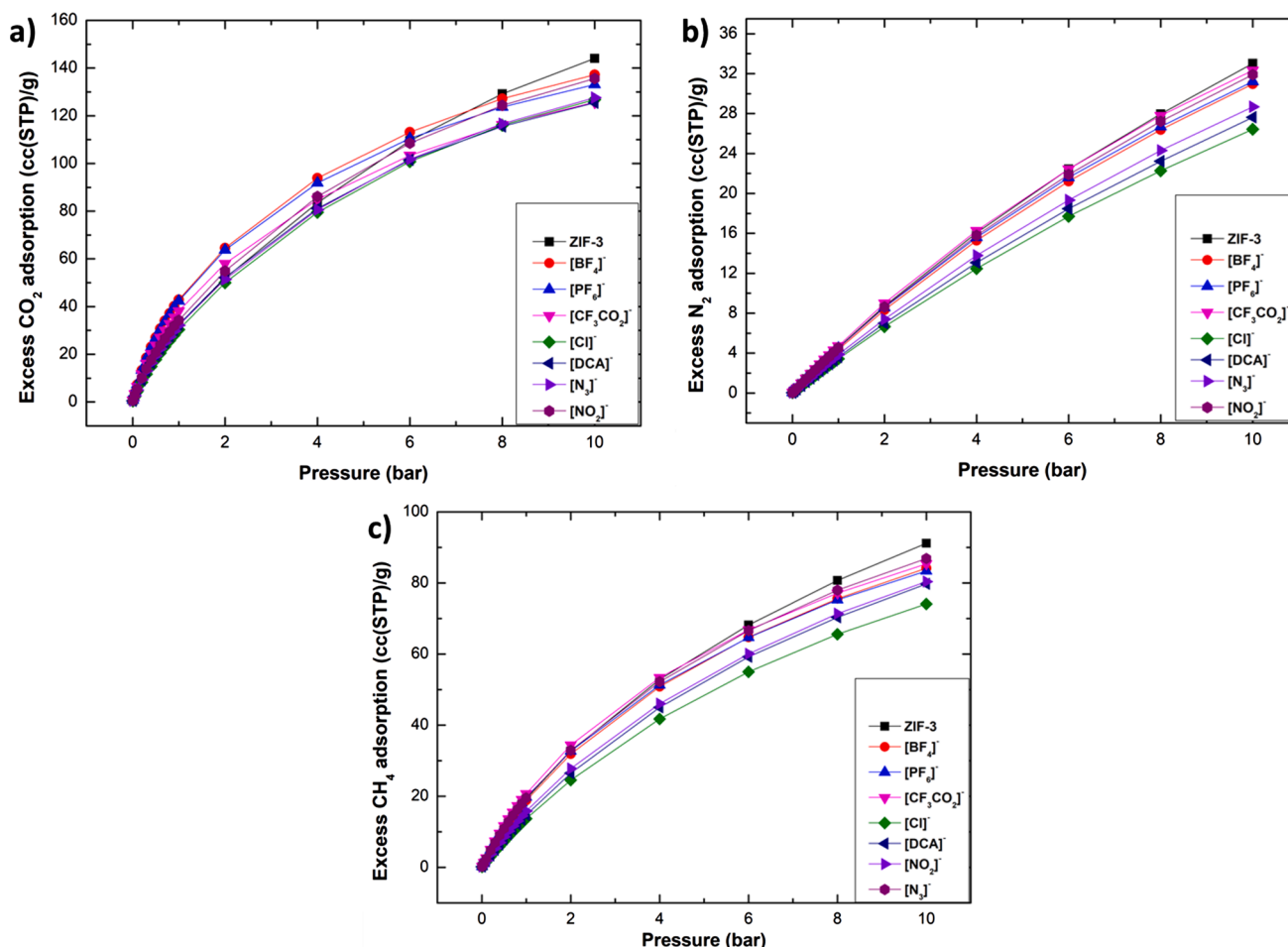


Fig. 3. Adsorption isotherm of different gases on ZIF-3 and ILs@ZIF-3 composites a) CO_2 , b) N_2 and c) CH_4 .

of ZIF-3 crossover that of $[\text{BMIM}][\text{BF}_4]@\text{ZIF-3}$ composite above 7 bar. But in binary adsorption, the composite shows higher adsorption than pristine ZIF-3, i.e. the composite selectively adsorb more CO_2 molecules from the mixture (CO_2/N_2) than ZIF-3. The designed composite shows good potential by adsorbing more CO_2 molecules from the gaseous mixture.

From the binary gas mixture adsorption, selectivity is calculated using Eq. (2). The selectivities of the gaseous mixtures are plotted against pressure and are given in Fig. 4. The designed material shows high CO_2/N_2 selectivity compared to other gaseous mixtures. Especially the $[\text{BF}_4]^-$ and $[\text{PF}_6]^-$ based composite materials show good CO_2 selectivity (Fig. 4a). A similar kind of behavior also exist in ILs@ZIF-8 as well [32,33]. The fluorine content in the anion can enhance the selective separation of CO_2 . The pristine ZIF-3 itself shows good CO_2/N_2 selectivity, which is further enhanced by the incorporation of IL. The non-fluorinated anions such as $[\text{Cl}]^-$, $[\text{DCA}]^-$, $[\text{N}_3]^-$ and $[\text{NO}_2]^-$ shows less selectivity when compared to the fluorinated ones. As the pressure increases, the selectivity is decreases as the gas molecules are completely occupied in the pores thus blocking further adsorption.

Fig. 4b and c gives the selectivities of CO_2/CH_4 and CH_4/N_2 . The CO_2 selectivity from CO_2/CH_4 mixture is ~ 4.5 times lesser compared to CO_2/N_2 mixture. This proves that the designed material is more suitable for industrial flue gas separation than geological and natural gas separation. It is important to note here that, fluorinated anions in IL shows better selectivity compared to other composites. It is evident from Fig. 4c that some of the composites shows low selectivity compared to pristine ZIF-3. The selectivity of $[\text{NO}_2]^-$, $[\text{DCA}]^-$ and $[\text{Cl}]^-$ based composites lies below that of ZIF-3. From the above discussions, it is clear that ZIF-3 enhances the selective CO_2 adsorption with the presence of IL in the channel pore.

ZIF-3 shows better selectivity compared to extensively studied ZIF-8. The comparison of CO_2/N_2 selectivities in pristine ZIFs (ZIF-3 and ZIF-8 [32]) and $[\text{BMIM}][\text{BF}_4]@\text{ZIFs}$ composites are given in Fig. 5. ZIF-3 shows better CO_2 selectivity than ZIF-8 at all the pressures. As the pressure increases, the difference between them is narrowed. At low pressure, $[\text{BMIM}][\text{BF}_4]@\text{ZIF-8}$ shows high CO_2 selectivity compared to $[\text{BMIM}][\text{BF}_4]@\text{ZIF-3}$. But as the pressure slightly increases, the selectivity of $[\text{BMIM}][\text{BF}_4]@\text{ZIF-8}$ goes under the selectivity of $[\text{BMIM}][\text{BF}_4]@\text{ZIF-3}$ (~ 0.5 bar). The newly predicted ILs@ZIF-3 can outperform the well-known ILs@ZIF-8 at elevated pressures. This happened due to the flexibility of linker at the elevated pressure. A similar finding was also observed from DFT calculations. The flexible nature of framework is highly influencing the CO_2 selectivity and this can be applicable in many industrial based separations. The selectivity of these materials can be further tuned by the selection of appropriate IL anions and functionalization of the framework.

3.2.3. Heat of adsorption

The ΔH_{ads} is and indicator of the interaction strength between the guest (gas) molecules and surface [56]. The heat of adsorption for different gas molecule on the ZIF-8, [32] ZIF-3 and ILs@ZIF-3 are given in Table 2. The ΔH_{ads} can be directly related to the selectivities shown by the complexes. Cao and co-workers predicted that difference of isosteric heat (DIH) can be effectively used for the screening of porous materials, because at zero pressure, the selectivity depends on the DIH [57]. When we compare ΔH_{ads} of different gases on ZIF-3 and ZIF-8, all three gas molecules shows enhancement in ΔH_{ads} in ZIF-3. This means that the ZIF-3 framework act as good material for gas storage and separation. The ΔH_{ads} of CO_2 in ZIF-3 is -23.24 kJ/mol. The same for ZIF-8 for CO_2

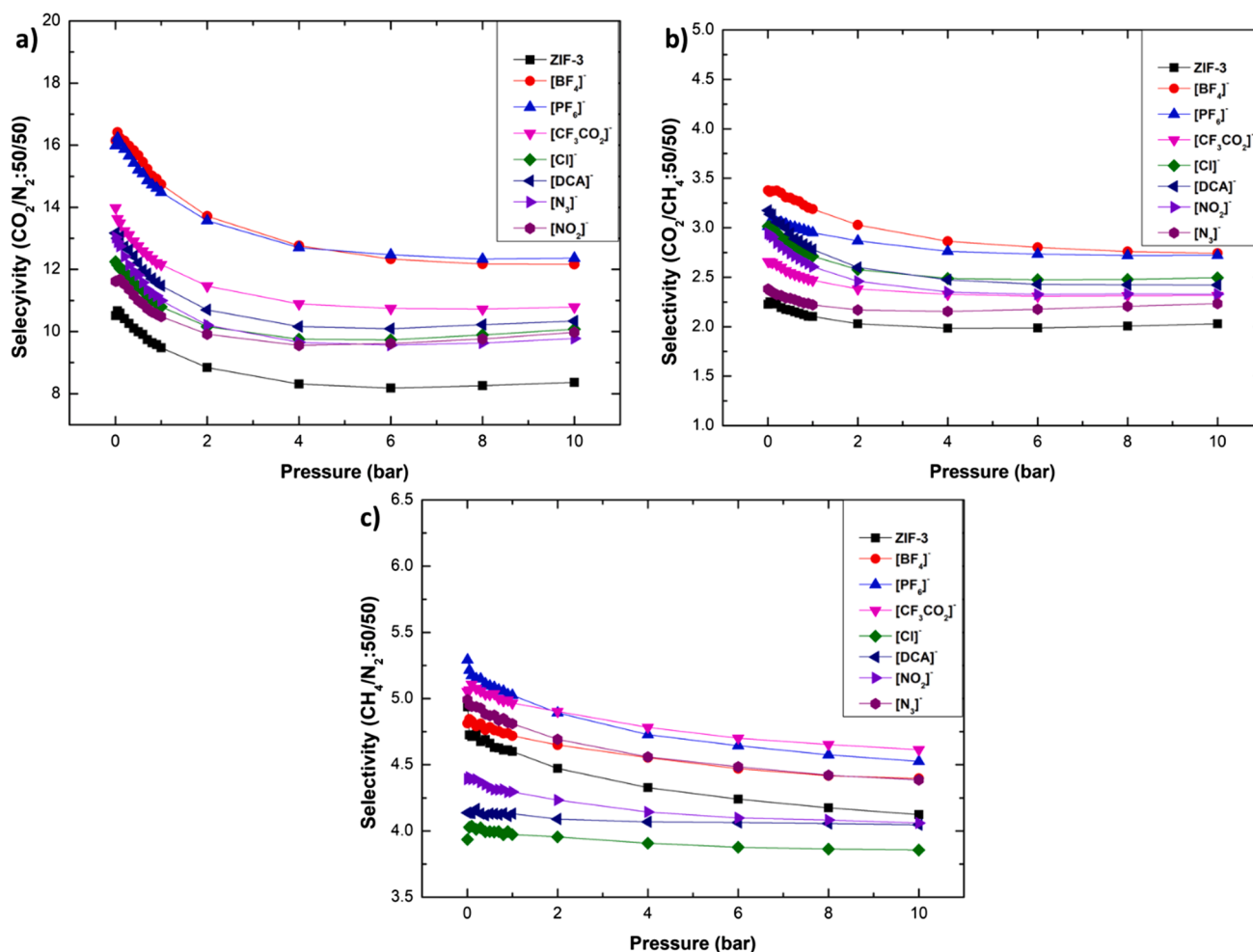


Fig. 4. Selectivities of different gaseous mixtures on ZIF-3 and ILs@ZIF-3 a) CO_2/N_2 , b) CO_2/CH_4 and c) CH_4/N_2 .

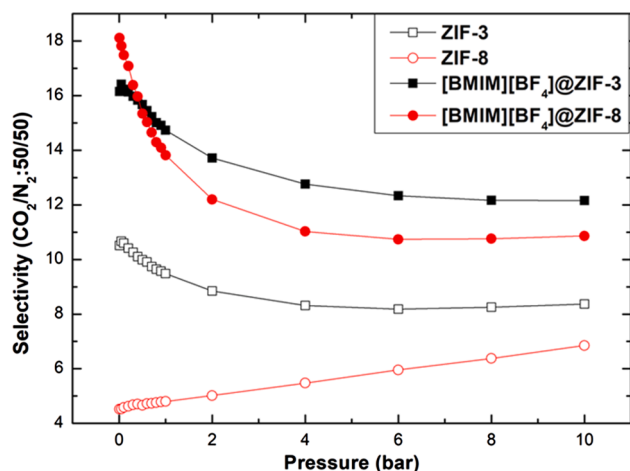


Fig. 5. Comparison of CO_2/N_2 selectivities of ZIF-3 and ZIF-8 (open symbol), and [BMIM][BF_4]@ZIF-3 and [BMIM][BF_4]@ZIF-8 (closed symbol), respectively.

is -15.38 kJ/mol. The calculated ΔH_{ads} clearly reveals that ZIF-3 has higher CO_2 affinity than ZIF-8. It is found that after the incorporation of IL at ZIF-3 there is a slight enhancement in the ΔH_{ads} . The ΔH_{ads} for [BMIM] $^+$ with [BF_4] $^-$ and [PF_6] $^-$ complexes are same (i.e. ~ -25.6 kJ/mol). Whereas other complexes show significantly less ΔH_{ads} value. In addition, we have also computed ΔH_{ads} for other two gases (i.e. CH_4 and

Table 2

The heat of adsorption of different gases on ZIF-3 and IL@ZIF-3 composites.

ZIF-3 and [BMIM] $^+$ [X] $^-$ @ZIF-3	Heat of adsorption (kJ/mol)		
	CO_2	N_2	CH_4
ZIF-8 ³²	-15.38	-10.19	-13.43
ZIF-3	-23.24	-13.35	-18.78
[BF_4] $^-$	-25.57	-14.07	-19.25
[PF_6] $^-$	-25.57	-14.39	-19.88
[CF_3CO_2] $^-$	-24.74	-14.27	-19.52
[Cl] $^-$	-24.27	-12.93	-17.39
[DCA] $^-$	-24.88	-13.29	-17.82
[N_3] $^-$	-24.72	-13.43	-18.31
[NO_2] $^-$	-23.86	-13.85	-19.19

N_2) which is less than the CO_2 . So the selected composite material act as potential membrane for CO_2 capture and it have the ability to separate gas mixtures of industrial applications. The N_2 gas has the lowest ΔH_{ads} . It is interesting to note that the ΔH_{ads} of CH_4 in [Cl] $^-$, [DCA] $^-$ and [N_3] $^-$ composites are less than that of pristine ZIF-3. These composites also showed lesser selectivity for CH_4/N_2 mixture than ZIF-3 (Fig. 4c). They do not show much affinity towards methane gas, which is evident from the ΔH_{ads} . The [Cl] $^-$ based composite also shows less ΔH_{ads} than ZIF-3 for N_2 . Thus it is clear that IL incorporation not compulsorily enhances the selectivity of the ZIF. They can perform better than the IL incorporated composites sometimes.

The enhancement in selectivity and ΔH_{ads} after [BMIM][BF_4] incorporation is visible from the geometries obtained after gas

adsorption (CO_2/N_2) from GCMC simulations. They are given in Fig. S8. Form the comparison of loading at ZIF-3 and the composite, it is evident that with IL incorporation, the CO_2 molecules are adsorbed more abundantly. At 1 bar pressure, the CO_2 molecules are very less in the ZIF-3 pore. But as the pressure goes up to 10 bar, more CO_2 and N_2 is entering the pore. When IL is already exist in the pore, the incoming gas molecule can interact with the ion-pair, especially with the anion through various kinds of non-covalent interactions [32,33]. At low pressure, as the number of incoming molecules are less, they form long-range van der Waals (vdWs) interactions with the ion-pair (~ 3.0 to 3.4 Å). At pressure 10 bar, more CO_2 is occupying the pore along with the IL ion-pair thus leads to halogen bonded interaction with the C of CO_2 and F of $[\text{BF}_4]^-$. The interacting distance varies from ~ 2.6 to 3 Å. This is the reason behind the enhancement in the ΔH_{ads} of CO_2 in the IL based composites thus leading to the enhancement in the selectivity. Also, the GCMC snapshots from CO_2/N_2 binary mixture adsorption at 10 bar pressure, for all the composites are given in Supporting Information Fig. S9 along with shortest interaction distances. In the $[\text{BF}_4]^-$ complex, the incoming CO_2 is mainly interacting with the anion of the IL at 2.65 Å. The cation does not have strong interaction with CO_2 . In case of $[\text{PF}_6]^-$ complex, the CO_2 interact with both anion and cation of the IL, with prominent interaction with the anion. CO_2 forms a weak H-bond with the cation at 2.98 Å. In $[\text{CF}_3\text{CO}_2]^-$ complex, the anion and cation have strong H-bond around 1.82 Å which leads reduced interaction with the incoming CO_2 and thus reduces the ΔH_{ads} compared to other two fluorinated anionic complexes. Similarly, for $[\text{Cl}]^-$, the strong interaction between anion and cation (2.05 Å) can weaken the interaction between composite and CO_2 . In case of $[\text{DCA}]^-$, butyl chain of cation is more strongly interacting with CO_2 (2.78 Å) than the anion (2.93 Å). In $[\text{N}_3]^-$ complex, the anion and cation has strong interaction at 1.79 Å and thus the CO_2 is more concentrated around the ZIF-3 framework than the IL. In $[\text{NO}_2]^-$, the N_2 gas form H-bond with butyl chain of cation at 2.83 Å. Also, the anion and cation has a stronger H-bond at 1.85 Å. These strong interactions decrease the interaction between composite and CO_2 molecule. This explains the lowest ΔH_{ads} observed for $[\text{NO}_2]^-$ complex. From this analysis, it is clear that the incoming CO_2 has strong interaction with IL and framework in all the composites.

Close analysis reveals that the gas capture performance of ZIF-3 is similar to well-known ZIF-8 membrane, but the composite stability significantly higher due to the interfacial interactions between IL and ZIF-3 surface [41]. This study can provide valuable information for the design of novel composite materials for the selective gas capture and separation applications.

4. Conclusions

The IL incorporation into the framework of ZIF-3 and resulting enhancement in the CO_2 separation is studied. ZIF-3 framework has a channel-type pore and high surface area than many of the frequently studied ZIFs. Hydrophobic and hydrophilic ILs were selected for the composite material and orientation effect is also considered in this study. The interfacial interactions and respective adsorption energies of IL@ZIF-3 complexes were quantified using periodic DFT calculations. Orientation with highest adsorption energy is further analyzed for CO_2 adsorption and separation using GCMC simulations. It was found that all ILs are compatible with ZIF-3 unlike in case of ZIF-8 (where nitrogenous anions are not compatible) [32]. It is due to the reactive $\text{C}_2\text{-H}$ in imidazole linker and different topology of ZIF-3. This is an important finding in identifying suitable ILs at the confinement with high stability.

This ZIF can adsorb more gas molecules than composites at high pressures (>7 bar). The calculated selectivities of different gaseous mixtures reveals that the ZIF-3 itself has good selectivity towards CO_2 . It is further improved with IL incorporation. The fluorinated anionic ILs shows more selectivity compared to other ILs. IL@ZIF-3 composite can perform better than the well-known IL@ZIF-8 composite at high pressure. ZIF-3 can perform better than some composites in selective

adsorption of CH_4 from N_2 . The ΔH_{ads} gives quantitative insight to the selectivity of the composites. With IL in the framework, the incoming CO_2 molecules form halogen bonds with the anion of IL and thus the ΔH_{ads} enhanced for fluorinated IL based composites. Studies of IL@ZIFs composite are limited to only a few ZIFs, currently. The evaluation of various ZIFs with appropriate pore size and surface area and IL impregnation into its pores can lead to the development of better materials for membrane based separation of various gas mixtures.

CRediT authorship contribution statement

Anoopa Thomas: Resources, Methodology, Investigation, Writing - original draft. **Muthuramalingam Prakash:** Conceptualization, Resources, Investigation, Writing - original draft.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

Authors thank the Department of Science and Technology-Science and Engineering Research Board (DST-SERB) of India for the financial support (Grant number: ECR/2017/000891). The authors also thank SRM-IST for providing the supercomputing facility and financial support.

Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.apsusc.2021.150173>.

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